

Anan Rajesh

Edinburgh, UK | +44 7470219561 | A.rajesh@sms.ed.ac.uk

<https://jithu04.github.io/x4tal/> | <https://www.linkedin.com/in/an-an-rajesh/>

3rd year transitioning to 4th year biomedical sciences undergraduate at the **University of Edinburgh**. I have optimised enzyme extraction, and made an initial discovery on the GBA gene in *C. elegans* (Gaucher-Parkinson's link). I am currently running an iGEM project (pharmaceutical toxicity biosensor) as a dry-lab team lead. I'm looking to pursue doctoral studies where I can apply my multidisciplinary approach to understand complex biological processes and develop therapies.

Primary Research Experience

5-Month Research Intern, Doitsidou Lab

Institute for Neuroscience and Cardiovascular Research, Edinburgh

Jul 2025 – Nov 2025

- **Optimised and established a novel protein extraction protocol** for lysing *C. elegans* model organisms while preserving high levels of enzyme activity.
- Designed a data analysis automation template in MS Excel and R for protein and enzyme quantification.
- **Discovered the coding subunit of the *C. elegans* GBA gene** (encoding glucocerebrosidase). This foundational work was subsequently characterised by a Master's student for a project on the Gaucher-Parkinson's link. **Protocol under review for publication (Jun 2026)**.
- Executed fluorescence microscopy, worm behavioural assays, genotyping, and cryopreservation while successfully replicating complex published methodologies.
- Maintained meticulous documentation of instrumentation use and experiments under GLP guidelines, while owning responsibility for stock management and onboarding/training an incoming master's student.

iGEM 2026 representing the University of Edinburgh

Team CondenseSense

Mar 2026 – Current

- Co-developing a synthetic biology pharmaceutical toxicity biosensor using liquid-liquid phase separation (LLPS) within a competitive 15-member undergraduate research team.
- Serving as Dry-Lab Analysis Lead, deploying analytical toolsets including QuPath, GraphPad PRISM, and ImageJ to build workflows.
- Contacted and coordinating an international network of advisory experts from the LLPS field, while managing wet-lab methodological verification
- Managing public engagement and digital infrastructure, including website maintenance, open-source technical documentation, and social media outreach.
- <https://condensesense.github.io/iGEM/> & <https://www.instagram.com/condense.sense/>

Team Principal Investigator, and Research Member

Edinburgh Scientific Research Association

Sept 2023 – Current

- **Year 3:** student Principal Investigator leading a 6-member team to evaluate the antimicrobial properties of caffeine on a nasal microbiome model, awarded a **£200 research grant**. Project formally recognised by Professor Debby Bogaert (Centre for Inflammation Research).
- **Year 2:** Executed literature mapping and data synthesis for a structured scoping review investigating the regulatory mechanisms of microRNAs along the gut-brain axis.
- **Year 1:** Collaborated within a 6-member research team to analyze the physiological impacts of cortisol on *Streptococcus salivarius*, mapping stress-induced alterations in oral health dynamics. Successfully secured a competitive **£641 student research grant**.

Work Experiences, Competitions, and Projects

Biology Department Head

Edinburgh Scientific Research Association

May 2025 – Current

- Recruited and guided **98** society members in developing research proposals and applying for grants, working alongside as a team member. Led meetings and workshops weekly as the main facilitator. Managed lab space allocations and curated a list of **5 projects based on research gaps** for new society members. Managed a fund pool of **£6000** for grants, lab bookings, consumables and reagents for the ESRA research groups.

Biohackathon Scotland

Team Wavio

Mar 2026

- Co-developed a biohackathon concept (team of 6) using cymatics to model Glioblastoma tissue in microgravity, backed by physics-informed simulations and CAD. Contributed biological data verification, evidence gathering, and presentation design. Competed against 100 PhD's and postdocs.

Bioinformatics Research Intern

BioNome

May 2024 – Aug 2024

- Executed computational workflows including molecular docking, molecular dynamics simulations (MDS), and ADMET screening (SwissADME) to evaluate target phytochemicals against 5-alpha-reductase. Generated quantitative data models and drafted technical manuscript sections.

Kitchen Member

Dobbies Garden Centre, Edinburgh

Nov 2023 – Current

- Demonstrated time-management and stamina by consistently balancing 20-hour part-time employment alongside a full-time honours curriculum and extracurricular research commitments.

Education

BSc. (Hons) Biomedical Sciences

The University of Edinburgh

Sep 2023 – May 2027

- Predicted Classification: First Class Honours; **12 Distinctions** (A) and **4 Upper 2:1** (B)
- Key Modules: Neuroscience, Pharmacology, Applied Pharmacology, Mechanisms of Brain Development, Microorganisms, Infection and Immunity

Academic Engagement and Certificates

- Biomed Peer-Assisted Learning Scheme leader (5/25 - current)
- Medical article editor (8/24 – 1/26)
- Immunology summer school (9/25)
- R for Biochemists 201 (1/25)
- BioZeen Mammalian Cell Workshop (6/24)
- 1st year representative, Biomedical Teaching Organisation (9/23 - 5/24)
- 1st year representative, Edinburgh University Biomedical Society (9/23 - 5/24)